

20251105 Derivation of Metallicity Perturbation Without Equilibrium Assumption

Starting Equations

From the bathtub model, we have:

Gas surface density evolution:

$$\frac{d\Sigma_g}{dt} = \Sigma_{\text{in}} - (1 - R + \lambda)\Sigma_{\text{SFR}}$$

Metal surface density evolution:

$$\frac{d\Sigma_Z}{dt} = y(1 - R)\Sigma_{\text{SFR}} - (1 - R + \lambda)\Sigma_{\text{SFR}}Z + \Sigma_{\text{in}}Z_{\text{in}}$$

Metallicity definition:

$$Z = \frac{\Sigma_Z}{\Sigma_g}$$

Step 1: Derive the Metallicity Evolution Equation

Using the product rule on $\Sigma_Z = Z\Sigma_g$:

$$\frac{d\Sigma_Z}{dt} = Z\frac{d\Sigma_g}{dt} + \Sigma_g\frac{dZ}{dt}$$

Substitute the gas evolution equation:

$$\frac{d\Sigma_Z}{dt} = Z[\Sigma_{\text{in}} - (1 - R + \lambda)\Sigma_{\text{SFR}}] + \Sigma_g\frac{dZ}{dt}$$

Equate with the metal evolution equation:

$$Z\Sigma_{\text{in}} - Z(1 - R + \lambda)\Sigma_{\text{SFR}} + \Sigma_g\frac{dZ}{dt} = y(1 - R)\Sigma_{\text{SFR}} - (1 - R + \lambda)\Sigma_{\text{SFR}}Z + \Sigma_{\text{in}}Z_{\text{in}}$$

Simplify (the $-Z(1 - R + \lambda)\Sigma_{\text{SFR}}$ terms cancel):

$$\Sigma_g \frac{dZ}{dt} = y(1 - R)\Sigma_{\text{SFR}} + (Z_{\text{in}} - Z)\Sigma_{\text{in}}$$

Therefore:

$$\frac{dZ}{dt} = \frac{1}{\Sigma_g} [y(1 - R)\Sigma_{\text{SFR}} + (Z_{\text{in}} - Z)\Sigma_{\text{in}}]$$

Step 2: Solve for Z Including Time Derivative

Rearrange the equation:

$$\Sigma_g \frac{dZ}{dt} = y(1 - R)\Sigma_{\text{SFR}} + (Z_{\text{in}} - Z)\Sigma_{\text{in}}$$

$$\Sigma_g \frac{dZ}{dt} = y(1 - R)\Sigma_{\text{SFR}} + Z_{\text{in}}\Sigma_{\text{in}} - Z\Sigma_{\text{in}}$$

$$\Sigma_g \frac{dZ}{dt} + Z\Sigma_{\text{in}} = y(1 - R)\Sigma_{\text{SFR}} + Z_{\text{in}}\Sigma_{\text{in}}$$

Solve for Z:

$$Z = \frac{y(1-R)\Sigma_{\text{SFR}} + Z_{\text{in}}\Sigma_{\text{in}} - \Sigma_g \frac{dZ}{dt}}{\Sigma_{\text{in}}}$$

$$Z = Z_{\text{in}} + y(1 - R)\frac{\Sigma_{\text{SFR}}}{\Sigma_{\text{in}}} - \frac{\Sigma_g}{\Sigma_{\text{in}}} \frac{dZ}{dt}$$

Step 3: Define the Quasi-Equilibrium Metallicity

Define Z_{eq} as the metallicity that would exist if $\frac{dZ}{dt} = 0$:

$$Z_{\text{eq}} = Z_{\text{in}} + y(1 - R)\frac{\Sigma_{\text{SFR}}}{\Sigma_{\text{in}}}$$

Then the actual metallicity can be written as:

$$Z = Z_{\text{eq}} - \frac{\Sigma_g}{\Sigma_{\text{in}}} \frac{dZ}{dt}$$

This shows that Z deviates from Z_{eq} by a correction term proportional to $\frac{dZ}{dt}$.

Step 4: Estimate the Metallicity Relaxation Timescale

From the metallicity evolution equation:

$$\frac{dZ}{dt} = \frac{1}{\Sigma_g} [y(1 - R)\Sigma_{\text{SFR}} + (Z_{\text{in}} - Z)\Sigma_{\text{in}}]$$

Consider a perturbation δZ from quasi-equilibrium. Near Z_{eq} :

$$\frac{d(\delta Z)}{dt} = \frac{1}{\Sigma_g} [y(1 - R)\delta\Sigma_{\text{SFR}} + (Z_{\text{in}} - Z_{\text{eq}} - \delta Z)\delta\Sigma_{\text{in}} - \delta Z\Sigma_{\text{in}}]$$

Since $Z_{\text{in}} - Z_{\text{eq}} = -y(1 - R)\frac{\Sigma_{\text{SFR}}}{\Sigma_{\text{in}}}$, and assuming variations are slow:

$$\frac{d(\delta Z)}{dt} \approx -\frac{\Sigma_{\text{in}}}{\Sigma_g} \delta Z + \text{driving terms}$$

The characteristic relaxation timescale for metallicity perturbations is:

$$\tau_Z = \frac{\Sigma_g}{\Sigma_{\text{in}}}$$

This is the **gas depletion timescale** (assuming Σ_{in} is the dominant source).

Step 5: Timescale Separation Condition

Let τ_{var} be the timescale over which Σ_{SFR} and Σ_{in} vary significantly.

Condition for quasi-equilibrium: $\tau_Z \ll \tau_{\text{var}}$

This means:

$$\frac{\Sigma_g}{\Sigma_{\text{in}}} \ll \tau_{\text{var}}$$

When this condition is satisfied, the correction term in Step 3 becomes negligible:

$$\left| \frac{\Sigma_g}{\Sigma_{\text{in}}} \frac{dZ}{dt} \right| \ll |Z_{\text{eq}} - Z_{\text{in}}|$$

Therefore:

$$Z \approx Z_{\text{eq}} = Z_{\text{in}} + y(1 - R)\frac{\Sigma_{\text{SFR}}}{\Sigma_{\text{in}}}$$

Step 6: Derive the Perturbation Relationship

Starting from the quasi-equilibrium relation:

$$Z \approx Z_{\text{in}} + y(1 - R) \frac{\Sigma_{\text{SFR}}}{\Sigma_{\text{in}}}$$

Consider small perturbations around a reference state (denoted by subscript 0):

$$Z_0 + \delta Z = Z_{\text{in}} + y(1 - R) \frac{\Sigma_{\text{SFR},0} + \delta \Sigma_{\text{SFR}}}{\Sigma_{\text{in},0} + \delta \Sigma_{\text{in}}}$$

Using the approximation $\frac{A+\delta A}{B+\delta B} \approx \frac{A}{B} \left(1 + \frac{\delta A}{A} - \frac{\delta B}{B}\right)$:

$$\delta Z \approx y(1 - R) \frac{\Sigma_{\text{SFR},0}}{\Sigma_{\text{in},0}} \left(\frac{\delta \Sigma_{\text{SFR}}}{\Sigma_{\text{SFR},0}} - \frac{\delta \Sigma_{\text{in}}}{\Sigma_{\text{in},0}} \right)$$

Step 7: Convert to Logarithmic Perturbations

For logarithmic perturbations:

$$\delta \log_{10}(Z) = \frac{1}{\ln 10} \frac{\delta Z}{Z_0}$$

Substitute the expression for δZ :

$$\delta \log_{10}(Z) = \frac{1}{\ln 10} \frac{y(1-R) \frac{\Sigma_{\text{SFR},0}}{\Sigma_{\text{in},0}}}{Z_0} \left(\frac{\delta \Sigma_{\text{SFR}}}{\Sigma_{\text{SFR},0}} - \frac{\delta \Sigma_{\text{in}}}{\Sigma_{\text{in},0}} \right)$$

Using $\frac{\delta X}{X} = \ln(10) \delta \log_{10}(X)$:

$$\delta \log_{10}(Z) = \frac{y(1-R) \frac{\Sigma_{\text{SFR},0}}{\Sigma_{\text{in},0}}}{Z_0} (\delta \log_{10}(\Sigma_{\text{SFR}}) - \delta \log_{10}(\Sigma_{\text{in}}))$$

From the quasi-equilibrium condition: $Z_0 = Z_{\text{in}} + y(1 - R) \frac{\Sigma_{\text{SFR},0}}{\Sigma_{\text{in},0}}$

Therefore:

$$\delta \log_{10}(Z) = \frac{Z_0 - Z_{\text{in}}}{Z_0} (\delta \log_{10}(\Sigma_{\text{SFR}}) - \delta \log_{10}(\Sigma_{\text{in}}))$$

Step 8: Final Result for Observable Metallicity

Since $12 + \log_{10}(\text{O}/\text{H}) = 8.69 + \log_{10}(Z/0.0139)$:

$$\delta[12 + \log_{10}(\text{O}/\text{H})] = \delta \log_{10}(Z)$$

Therefore:

$$\delta[12 + \log_{10}(\text{O}/\text{H})] = \frac{Z - Z_{\text{in}}}{Z} (\delta \log_{10}(\Sigma_{\text{SFR}}) - \delta \log_{10}(\Sigma_{\text{in}}))$$

Or more compactly:

$$\delta[12 + \log_{10}(\text{O}/\text{H})] \propto \delta \log_{10}(\Sigma_{\text{SFR}}) - \delta \log_{10}(\Sigma_{\text{in}})$$

Conclusion

Key Result: The relationship $\delta[12 + \log_{10}(\text{O}/\text{H})] \propto \delta \log_{10}(\Sigma_{\text{SFR}}) - \delta \log_{10}(\Sigma_{\text{in}})$ holds **without** assuming $\frac{dZ}{dt} = 0$, as long as the timescale separation condition $\tau_Z \ll \tau_{\text{var}}$ is satisfied.

Physical Interpretation: The metallicity "tracks" the quasi-equilibrium value because it adjusts to changes in Σ_{SFR} and Σ_{in} much faster than these quantities vary. This is a **quasi-static approximation** based on timescale separation, not true equilibrium.

Bonus: Demonstrating $\tau_Z \ll \tau_{\text{var}}$

Typical Values

For a star-forming galaxy:

- $\Sigma_g \sim 10 \text{ M}_{\odot} \text{ pc}^{-2}$
- $\Sigma_{\text{in}} \sim 1 \text{ M}_{\odot} \text{ pc}^{-2} \text{ Gyr}^{-1}$
- $\Sigma_{\text{SFR}} \sim 0.1 \text{ M}_{\odot} \text{ pc}^{-2} \text{ Gyr}^{-1}$

Metallicity Relaxation Timescale

$$\tau_Z = \frac{\Sigma_g}{\Sigma_{\text{in}}} = \frac{10 \text{ M}_{\odot} \text{ pc}^{-2}}{1 \text{ M}_{\odot} \text{ pc}^{-2} \text{ Gyr}^{-1}} = 10 \text{ Gyr}$$

SFR Variation Timescale

Typical variations in SFR occur on timescales of:

- **Secular evolution:** $\tau_{\text{var}} \sim \text{few Gyr}$ (Hubble time)
- **Galaxy interactions:** $\tau_{\text{var}} \sim 0.1 - 1 \text{ Gyr}$
- **Within simulation timesteps:** If you're resolving variations on $\sim 100 \text{ Myr}$ timescales

Wait, this doesn't satisfy $\tau_Z \ll \tau_{\text{var}}$!

Resolution: The Effective Timescale

The relevant comparison is not just $\frac{\Sigma_g}{\Sigma_{\text{in}}}$, but the actual metallicity adjustment rate including outflows:

From the full equation:

$$\frac{dZ}{dt} = \frac{1}{\Sigma_g} [y(1 - R)\Sigma_{\text{SFR}} + (Z_{\text{in}} - Z)\Sigma_{\text{in}}]$$

Near equilibrium, the restoring force is:

$$\frac{d(\delta Z)}{dt} \approx -\frac{\Sigma_{\text{in}}}{\Sigma_g} \delta Z$$

But the **total metal cycling rate** includes star formation:

$$\tau_Z^{\text{eff}} = \frac{\Sigma_g}{\Sigma_{\text{in}} + (1 - R + \lambda)\Sigma_{\text{SFR}}}$$

With $\Sigma_{\text{SFR}} \sim 0.1 \text{ M}_{\odot} \text{ pc}^{-2} \text{ yr}^{-1}$ and $(1 - R + \lambda) \sim 0.6$:

$$\tau_Z^{\text{eff}} = \frac{10}{1 + 0.6 \times 0.1} \approx 9.4 \text{ Gyr}$$

Why It Still Works

Even though $\tau_Z \sim \tau_{\text{var}}$, the quasi-equilibrium approximation works because:

1. **The metallicity evolution is a damped response:** Perturbations decay exponentially with e^{-t/τ_Z}
2. **Most astrophysical variations are gradual:** SFR doesn't jump discontinuously
3. **In practice:** If you're looking at correlations on timescales $> 0.1\tau_Z$, the metallicity has already adjusted significantly

Quantitative criterion: The approximation is good when:

$$\frac{1}{\tau_Z} \frac{d \ln \Sigma_{\text{SFR}}}{dt} \ll 1$$

This means the fractional change in Σ_{SFR} per metallicity relaxation time is small, which is typically true for secular evolution even when $\tau_Z \sim \tau_{\text{var}}$.